

40.2 Generalized linear mixed models

Replace the scalar U_i of [Section 40.1](#) by a vector of random effects with a covariance matrix, and the single covariate by a model matrix, and the result is the generalized linear mixed model: the linear mixed model of [Chapter 32](#) with a link function and an exponential dispersion family in place of the normal error.

That one change costs more than it looks. In Definition 32.2.1 the marginal distribution was still normal, with covariance $\mathbf{V} = \mathbf{ZGZ}^\top + \mathbf{R}$, so everything Part VII knew about generalized least squares applied unchanged. Here it is an integral with no closed form, and the marginal moments have to be computed one family at a time.

40.2.1 The model

Definition 40.2.1 — Generalized linear mixed model

Let $\mathbf{u} \sim N_q\{\mathbf{0}, \mathbf{G}(\boldsymbol{\theta})\}$ be a vector of **random effects** with covariance depending on unknown variance components $\boldsymbol{\theta}$. Given $\mathbf{u}$, let the responses $Y_1, \dots, Y_n$ be independent, each from the exponential dispersion family Definition 34.1.1 with prior weight w_i , dispersion ϕ and variance function V , and with conditional mean

$$\mu_i(\mathbf{u}) = E(Y_i | \mathbf{u}) = h(\eta_i), \quad \eta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \mathbf{z}_{(i)}^\top \mathbf{u}, \quad (40.2.1)$$

so that $\text{Var}(Y_i | \mathbf{u}) = \phi V\{\mu_i(\mathbf{u})\}/w_i$. In matrix form $\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}$ with $\mathbf{X}$ of size $n \times p$ and $\mathbf{Z}$ of size $n \times q$, both known. The parameters are the **fixed effects** $\boldsymbol{\beta}$, the variance components $\boldsymbol{\theta}$ and, if it is not fixed by the family, the dispersion ϕ .

Setting $\mathbf{Z} = \mathbf{0}$ recovers the generalized linear model Definition 34.2.1 exactly; taking the family normal and h the identity, with $\mathbf{R} = \phi\mathbf{I}$, recovers Definition 32.2.1. Two structures cover most applications.

Clustered data. Units $i = 1, \dots, m$ with n_i observations each, one random effect vector $\mathbf{u}_i \sim N_{q_0}(\mathbf{0}, \mathbf{G}_0)$ per unit, independent across units. Then $\mathbf{Z}$ is block diagonal, $q = mq_0$, and both the likelihood and the marginal covariance break into m independent pieces. The **random-intercept** model takes $q_0 = 1$ and $\mathbf{z}_{(ij)} = 1$:

$$g(\mu_{ij}) = \mathbf{x}_{(ij)}^\top \boldsymbol{\beta} + u_i, \quad u_i \stackrel{\text{iid}}{\sim} N(0, \tau^2). \quad (40.2.2)$$

Crossed or nested effects. Pupils in classes in schools; raters and items. Here $\mathbf{G}$ is block diagonal in the *effects* rather than the units and the likelihood does not factorize. [Chapter 32](#) described the linear versions (Example (Nested and crossed random factors)); the difficulty a link function adds is computational, and severe, because the integral in [Section 40.3](#) is then genuinely high-dimensional.

Remark (Two extensions, and why they are rarer). A cluster-level random effect can be put into the linear predictor of a baseline-category model Definition 36.1.1 or of a cumulative-logit model Definition 36.2.1, giving mixed models for nominal and ordinal responses; in the ordinal case the effect shifts every cutpoint together, leaving

the proportional-odds structure intact. Conditional independence given $\mathbf{u}_i$ can also be relaxed, by letting the random effects themselves follow an AR(1) process within the cluster — the mixed-model counterpart of the non-diagonal working correlation of [Section 40.5](#). That is rare, because it raises each cluster's integral from dimension q_0 to n_i .

Example. (A random-intercept logistic model)

m physicians each treat n_i patients; $Y_{ij} = 1$ if patient j of physician i recovers. With the logistic link of Definition 35.1.1, $\text{logit Pr}(Y_{ij} = 1 \mid u_i) = \mathbf{x}_{(ij)}^\top \boldsymbol{\beta} + u_i$ and $u_i \sim N(0, \tau^2)$, the coefficient of a patient-level covariate compares two patients *of the same physician*, and τ measures how much physicians differ. A useful summary is the **median odds ratio** $\exp(\sqrt{2} \tau z_{0.75})$, the median odds ratio between two randomly chosen physicians treating identical patients ([Exercise \(B1\)](#)). Replacing u_i by $u_{0i} + u_{1i}x_{ij}$ gives a random-slope model, at the price of three parameters in $\mathbf{G}_0$ and, by [Exercise \(B2\)](#), no single marginal coefficient at all.

Example. (The Poisson–lognormal model)

With $Y_i \mid u_i$ Poisson (Definition 37.1.1) of mean $\exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta} + u_i)$ and $u_i \sim N(0, \tau^2)$, the conditional mean is multiplied by the lognormal variable e^{u_i} . Unlike the binary case this model is identified with a *single* observation per unit, because the Poisson variance is determined by its mean and any excess must come from u_i : it is an overdispersion model in the sense of [Chapter 38](#) (Proposition 38.4.1), and a direct competitor to the negative binomial of Definition 37.4.1, which arises the same way with a gamma mixing variable (Lemma 37.4.1).

40.2.2 The moments a random effect induces

Proposition 40.2.1 — Marginal moments of a GLMM

In [Definition 40.2.1](#) write $\mu_i(\mathbf{u}) = h(\eta_i)$ and $\mu_i^M = E\{\mu_i(\mathbf{u})\}$.

$$(a) \quad E(Y_i) = \mu_i^M,$$

$$\text{Var}(Y_i) = \frac{\phi}{w_i} E[V\{\mu_i(\mathbf{u})\}] + \text{Var}\{\mu_i(\mathbf{u})\}, \quad (40.2.3)$$

and $\text{Cov}(Y_i, Y_k) = \text{Cov}\{\mu_i(\mathbf{u}), \mu_k(\mathbf{u})\}$ for $i \neq k$. The marginal variance always exceeds the average conditional variance.

(b) **Binomial.** If $Y_i \mid \mathbf{u} \sim \text{Bin}\{N_i, \pi_i(\mathbf{u})\}$ and $\bar{\pi}_i = E \pi_i(\mathbf{u})$, then $E(Y_i) = N_i \bar{\pi}_i$ and

$$\text{Var}(Y_i) = N_i \bar{\pi}_i (1 - \bar{\pi}_i) \left\{ 1 + (N_i - 1) \frac{\text{Var}\{\pi_i(\mathbf{u})\}}{\bar{\pi}_i (1 - \bar{\pi}_i)} \right\}. \quad (40.2.4)$$

The overdispersion factor is the beta-binomial one, and it equals 1 when $N_i = 1$: a *single* Bernoulli response carries no trace of the random effect.

(c) **Poisson with a log link**, $\phi = 1$, and a random intercept $u_i \sim N(0, \tau^2)$ shared

by the observations of a cluster. Then $\mu_{ij}^M = \exp(\mathbf{x}_{(ij)}^\top \boldsymbol{\beta} + \tau^2/2)$,

$$\begin{aligned}\text{Var}(Y_{ij}) &= \mu_{ij}^M + (\mu_{ij}^M)^2 (e^{\tau^2} - 1), \\ \text{Cov}(Y_{ij}, Y_{ik}) &= \mu_{ij}^M \mu_{ik}^M (e^{\tau^2} - 1).\end{aligned}\tag{40.2.5}$$

The marginal variance function is *exactly* the NB2 function $V(\mu) = \mu + \mu^2/\kappa$ of Definition 37.4.1, with $\kappa = 1/(e^{\tau^2} - 1)$ — although the marginal distribution is not negative binomial and has no closed form.

Proof.

- (a) Condition on $\mathbf{u}$ and use the variance decomposition $\text{Var}(Y) = \text{E}\{\text{Var}(Y | \mathbf{u})\} + \text{Var}\{\text{E}(Y | \mathbf{u})\}$ and, for $i \neq k$, $\text{Cov}(Y_i, Y_k) = \text{E}\{\text{Cov}(Y_i, Y_k | \mathbf{u})\} + \text{Cov}\{\text{E}(Y_i | \mathbf{u}), \text{E}(Y_k | \mathbf{u})\}$, whose first term vanishes by conditional independence. The final claim is $\text{Var}\{\mu_i(\mathbf{u})\} \geq 0$.
- (b) With $\pi = \pi_i(\mathbf{u})$, Eq. 40.2.3 gives $\text{Var}(Y_i) = N_i \text{E}\{\pi(1 - \pi)\} + N_i^2 \text{Var}(\pi)$. Now $\text{E}\{\pi(1 - \pi)\} = \bar{\pi}_i - \text{E}(\pi^2) = \bar{\pi}_i(1 - \bar{\pi}_i) - \text{Var}(\pi)$, so $\text{Var}(Y_i) = N_i \bar{\pi}_i(1 - \bar{\pi}_i) + N_i(N_i - 1) \text{Var}(\pi)$, which is Eq. 40.2.4.
- (c) For a lognormal factor, $\text{E} e^{su} = e^{s^2 \tau^2/2}$. Hence $\mu_{ij}^M = e^{\eta_{ij}^0} \text{E} e^{u_i} = e^{\eta_{ij}^0 + \tau^2/2}$, writing $\eta_{ij}^0 = \mathbf{x}_{(ij)}^\top \boldsymbol{\beta}$. By Eq. 40.2.3 with $V(\mu) = \mu$ and $\phi = w_{ij} = 1$,

$$\text{Var}(Y_{ij}) = \mu_{ij}^M + e^{2\eta_{ij}^0} (\text{E} e^{2u_i} - (\text{E} e^{u_i})^2) = \mu_{ij}^M + e^{2\eta_{ij}^0} (e^{2\tau^2} - e^{\tau^2}),$$

and $e^{2\eta_{ij}^0} (e^{2\tau^2} - e^{\tau^2}) = (e^{\eta_{ij}^0 + \tau^2/2})^2 (e^{\tau^2} - 1)$. The covariance is the same computation with $\eta_{ij}^0 + \eta_{ik}^0$ in the exponent. Matching $\mu + \mu^2(e^{\tau^2} - 1)$ with $\mu + \mu^2/\kappa$ gives κ . ■

Part (b) explains why a random-intercept model for *ungrouped* binary data needs repeated observations: with $N_i = 1$ the marginal distribution is Bernoulli whatever τ is, and τ is identified only through the within-cluster *correlation*. Part (c) explains why the Poisson–lognormal and the negative binomial are so hard to tell apart: they agree in their first two moments exactly and differ only in shape, so a likelihood ratio between them compares tails, not overdispersion. The listing checks Eq. 40.2.5 by simulation at $\tau = 0.6$, where the matching NB2 shape is $\kappa = 1/(e^{0.36} - 1) = 2.3077$.

```
import numpy as np

rng = np.random.default_rng(40_002)
tau, eta1, eta2 = 0.6, 0.4, -0.2
u = rng.normal(0, tau, 1_000_000)
y1 = rng.poisson(np.exp(eta1 + u))
y2 = rng.poisson(np.exp(eta2 + u))
mu1, mu2 = np.exp(eta1 + tau**2 / 2), np.exp(eta2 + tau**2 / 2)

print(f"E Y1      : {y1.mean():.4f} (theory {mu1:.4f})")
print(f"Var Y1     : {y1.var():.4f} (theory {mu1 + mu1**2 * (np.exp(tau**2) - 1):.4f})")
print(f"Cov(Y1,Y2): {np.cov(y1, y2)[0, 1]:.4f} (theory {mu1 * mu2 * (np.exp(tau**2) - 1):.4f})")
```

[Run this code online](#)

40.2.3 What a random intercept cannot do

Lemma 40.2.2 — A shared random intercept induces nonnegative correlation

Let U be a random variable and $f, g : \mathbb{R} \rightarrow \mathbb{R}$ both nondecreasing, with $E f(U)^2$ and $E g(U)^2$ finite. Then $\text{Cov}\{f(U), g(U)\} \geq 0$. Consequently, in a random-intercept GLMM Eq. 40.2.2 with any inverse link h that is nondecreasing,

$$\text{Cov}(Y_{ij}, Y_{ik}) = \text{Cov}\{h(\eta_{ij}^0 + u_i), h(\eta_{ik}^0 + u_i)\} \geq 0 \quad (j \neq k).$$

Proof. Let U' be an independent copy of U . Because both functions are nondecreasing, $\{f(U) - f(U')\}\{g(U) - g(U')\} \geq 0$ pointwise. Taking expectations and expanding,

$$0 \leq E[\{f(U) - f(U')\}\{g(U) - g(U')\}] = 2[E\{f(U)g(U)\} - E f(U) E g(U)] = 2 \text{Cov}\{f(U), g(U)\}.$$

The second claim is Proposition 40.2.1(a) with $f(u) = h(\eta_{ij}^0 + u)$ and $g(u) = h(\eta_{ik}^0 + u)$, both nondecreasing. ■

Every link in this book has a nondecreasing inverse, so the consequence is a real restriction: **a model whose only random effect is a cluster-level intercept cannot describe negatively correlated responses.** Such data exist. If exactly one of a traveller's four options is chosen, or exactly one of a litter survives a fixed food supply, the responses compete and their correlation is negative by construction. Section 40.4 analyses a data set of that kind, and the estimating-equation approach has no difficulty with it precisely because it never commits to a mechanism for the dependence.

40.2.4 Estimation, inference and prediction

Fitting is the subject of Section 40.3. The random effects cannot be estimated by maximizing a joint likelihood, because their number grows with the sample size — the incidental-parameter trap of Example (Many means, one variance). They are integrated out, giving the **integrated likelihood**

$$L(\boldsymbol{\beta}, \boldsymbol{\theta}, \phi) = \int_{\mathbb{R}^q} \prod_{i=1}^n f(y_i | \mathbf{u}; \boldsymbol{\beta}, \phi) p(\mathbf{u}; \boldsymbol{\theta}) d\mathbf{u}, \quad (40.2.6)$$

which is maximized numerically. Inference for $\boldsymbol{\beta}$ is then the usual likelihood inference — Wald intervals from the inverse observed information, likelihood ratio tests between nested mean models — as $m \rightarrow \infty$ with cluster sizes bounded. That claim is imported, not proved here: Theorem 34.5.1 covers a generalized linear model, not a GLMM, and the conditions under which the integrated likelihood behaves like an ordinary one are set out by McCulloch, Searle and Neuhaus (2008, chapter 14). Two warnings carry over from Part VII unchanged. First, a test of $\tau^2 = 0$, or of any variance component on the boundary of its range, is *not* a $\chi^2(1)$ test: the limiting null distribution is the mixture $\frac{1}{2}\chi^2(0) + \frac{1}{2}\chi^2(1)$ under the conditions of Theorem 32.5.2, so the naive p -value is twice what it should be. Second, REML has no clean analogue, because

Definition 32.4.1 was built on the distribution of $\mathbf{K}^\top \mathbf{Y}$ for $\mathcal{C}(\mathbf{K}) = \mathcal{C}(\mathbf{X})^\perp$, a construction that needs the normal linear model; the approximate versions in use inherit the bias of the working linear model they are built on.

In the linear mixed model Theorem 32.3.1 produced a linear predictor of the random effects, and Eq. 32.3.3 showed it shrinks the cluster's own estimate towards zero. Neither linearity nor an explicit formula survives the link function, but the shrinkage does.

Proposition 40.2.3 — Empirical Bayes prediction and shrinkage

Consider Eq. 40.2.2 with $\boldsymbol{\beta}$, τ^2 and ϕ known, write $\ell_i(u) = \sum_j \log f(y_{ij} | u)$ for cluster i 's conditional log-likelihood, and assume ℓ_i is concave in u — as it is for a canonical link by Theorem 34.3.3, and for the gamma with a log link by direct differentiation.

- (a) The posterior density of u_i given $\mathbf{y}_i$, proportional to $\exp\{\ell_i(u) - u^2/(2\tau^2)\}$, is log-concave and its mode $\hat{u}_i$ is unique.
- (b) If ℓ_i attains its maximum at a finite $\hat{a}_i \neq 0$, then $\hat{u}_i$ lies strictly between 0 and $\hat{a}_i$: the prediction has the same sign as the cluster's own estimate and is strictly smaller in magnitude. If ℓ_i has no finite maximizer — as when every response in a binary cluster is 0 — then $\hat{u}_i$ is finite all the same.
- (c) Replacing ℓ_i by its quadratic expansion at $\hat{a}_i$ gives $\hat{u}_i \approx \{\tau^2/(\tau^2 + I_i^{-1})\} \hat{a}_i$ with $I_i = -\ell_i''(\hat{a}_i)$. In the two cases used in this chapter $I_i^{-1} = \phi/n_i$ exactly, which is Eq. 32.3.3: clusters with few observations are shrunk harder, and all clusters are shrunk harder when τ^2 is small relative to the noise.

Proof.

- (a) Adding the strictly concave $-u^2/(2\tau^2)$ to the concave ℓ_i keeps it concave and makes it strictly so, and the sum tends to $-\infty$ in both directions, so a unique maximizer exists.
- (b) Write $\psi(u) = \ell_i(u) - u^2/(2\tau^2)$ and suppose $\hat{a}_i > 0$. Then $\ell_i'(\hat{a}_i) = 0$ gives $\psi'(\hat{a}_i) = -\hat{a}_i/\tau^2 < 0$, while $\psi'(0) = \ell_i'(0) > 0$ because ℓ_i is concave with maximum to the right of 0. A continuous derivative positive at 0 and negative at $\hat{a}_i$ vanishes strictly between them, and by strict concavity that zero is the unique maximizer; the case $\hat{a}_i < 0$ is symmetric. If ℓ_i increases everywhere, ψ' is still eventually negative because of the penalty, so $\hat{u}_i$ is finite.
- (c) With $\ell_i(u) \approx \ell_i(\hat{a}_i) - \frac{1}{2}I_i(u - \hat{a}_i)^2$, the penalized maximizer is $\hat{u}_i = I_i\hat{a}_i/(I_i + \tau^{-2}) = \{\tau^2/(\tau^2 + I_i^{-1})\}\hat{a}_i$. For a canonical link $I_i = \sum_j w_{ij}V(\hat{\mu}_{ij})/\phi$, so $I_i^{-1} = \phi/(n_i\bar{V}_i)$ with $\bar{V}_i$ the average of the $w_{ij}V(\hat{\mu}_{ij})$; in the normal case $V \equiv 1$, $w \equiv 1$ and $\bar{V}_i = 1$. For the gamma with a log link $\ell_i''(u) = -\nu \sum_j y_{ij}e^{-\eta_{ij}}$ with $\nu = 1/\phi$, and $\ell_i'(\hat{a}_i) = 0$ forces $\sum_j y_{ij}e^{-\eta_{ij}} = n_i$ there, so again $I_i = n_i/\phi$ exactly.

■

Remark (The predictions are not estimates). $\hat{u}_i$ predicts a random variable, so its error is a prediction error variance and the interval covering u_i is a *prediction* interval (Theorem 12.3.1), exactly as in Theorem 32.3.1(b). The usual intervals ignore the estimation of $\boldsymbol{\beta}$ and $\boldsymbol{\theta}$ and are therefore too short when m is small.

40.2.5 An investment panel with a multiplicative mean

Example. (Grunfeld’s firms, again)

Chapter 32 fitted a linear mixed model to Grunfeld’s panel of 11 US firms observed in each of 20 years (Example (Grunfeld’s panel with a random firm effect)). Investment there ranges over three orders of magnitude and is far more variable for large firms than for small ones, which a constant-variance model cannot represent; a gamma response with a log link can, because the mean is multiplicative and the conditional coefficient of variation is constant. With Y_{ij} the gross investment of firm i in year j ,

$Y_{ij} \mid u_i \sim \text{Gamma}$ with $\log E(Y_{ij} \mid u_i) = \beta_0 + \beta_1 \log(\text{value}_{ij}) + \beta_2 \log(\text{capital}_{ij}) + u_i$, $u_i \sim N(0, \tau^2)$, and $\text{Var}(Y_{ij} \mid u_i) = \phi E(Y_{ij} \mid u_i)^2$. Fitting by adaptive Gauss–Hermite quadrature (Section 40.3) gives

	β_1 (value)	β_2 (capital)	ϕ	τ
marginal	0.8421	0.3386	0.2644	—
gamma GLM				
gamma GLMM	0.6825	0.2315	0.0930	0.5188

The firm-to-firm standard deviation $\hat{\tau} = 0.5188$ is large: firms a standard deviation apart on the log scale differ by a factor of $e^{0.5188} \approx 1.68$ in investment at the same value and capital stock. The conditional coefficient of variation is $\sqrt{\hat{\phi}} = 0.3049$. For a gamma with shape $1/\hat{\phi}$ the conditional variance of $\log Y$ is the trigamma function of the shape, 0.0974, so the fraction of the variance of $\log Y$ attributable to the firm is $\hat{\tau}^2/(\hat{\tau}^2 + 0.0974) = 0.734$.

The fitted conditional intercept is -1.3032 ; by Proposition 40.1.1(b) the log link causes no attenuation of the slopes, and the marginal intercept is $-1.3032 + \hat{\tau}^2/2 = -1.1686$. The difference between the two rows is therefore *not* the marginal–conditional distinction. It is the older within-versus-between problem of Section 6.6: the independence fit lets differences between firms contribute to the slopes, while the mixed model assigns most of them to u_i and estimates the slopes mainly from movement within a firm. A gap of this size accuses the mean model, and Section 40.5 returns to it with a diagnostic and two candidate explanations, Exercise (C1) with a remedy.

With 20 years per firm and $\hat{\phi} = 0.0930$ small there is almost nothing to shrink: the empirical slope of $\hat{u}_i$ on $\hat{a}_i$ is 0.9830, and Proposition 40.2.3(c) predicts 0.9830. Using only each firm’s first three years, the same parameters give 0.8951. Shrinkage measures how much a cluster’s own data are worth, and is not a fixed property of the model.

40.2.6 Exercises

40.2.6.1 A. Check your understanding

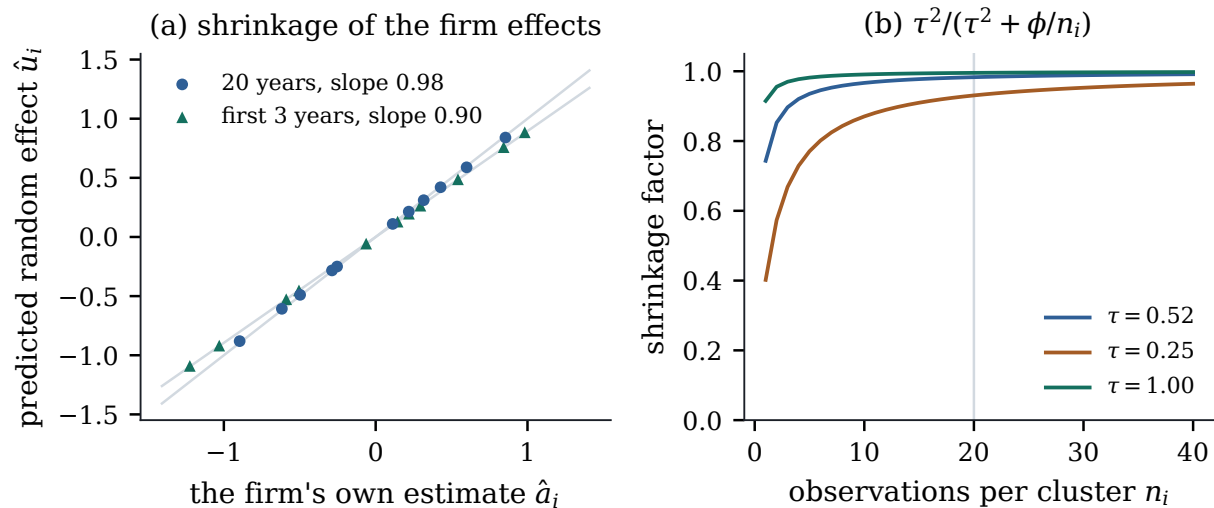


Figure 40.2.1: Shrinkage of the firm effects in the gamma GLMM of [Example \(Grunfeld's firms, again\)](#). (a) $\hat{u}_i$ against the firm's own estimate $\hat{a}_i$, for 20 years (circles) and for the first 3 only (triangles); the diagonal is no shrinkage. (b) The factor $\tau^2/(\tau^2 + \phi/n_i)$ of [Proposition 40.2.3\(c\)](#) at $\hat{\phi}$.

Exercise A1.

For each of the following, say whether the random-intercept variance τ^2 is identified, and why: (a) one Bernoulli response per cluster; (b) one binomial response with $N_i = 40$ trials per cluster; (c) one Poisson count per cluster; (d) five Bernoulli responses per cluster.

Exercise A2.

A Poisson–lognormal model is fitted and gives $\hat{\tau}^2 = 0.5$. What NB2 shape parameter κ would produce the same marginal variance function? Would you expect the two models to give noticeably different fitted means?

40.2.6.2 B. Practice

Exercise B1.

In [Example \(A random-intercept logistic model\)](#), let two physicians be drawn at random and let OR be the odds ratio between them for identical patients, taken in the direction that makes it at least one. Show that the median of OR is $\exp(\sqrt{2}\tau z_{0.75})$, where $z_{0.75} = 0.6745$ is the upper quartile of the standard normal. Evaluate it for $\tau = 1$.

Solution. The log odds ratio between physicians i and k is $u_i - u_k$, which is $N(0, 2\tau^2)$. Taking it in the positive direction gives $|u_i - u_k|$, whose median is the upper quartile of $N(0, 2\tau^2)$, namely $\sqrt{2}\tau z_{0.75}$. Exponentiating gives the claim. For $\tau = 1$, $\sqrt{2} \times 0.6745 = 0.9539$ and $e^{0.9539} \approx 2.6$: a typical pair of physicians differs by a factor of about 2.6 in the odds of recovery, for identical patients.

Exercise B2.

Give a two-level model with a *vector* random effect whose induced within-cluster correlation is negative, and explain why [Lemma 40.2.2](#) does not apply. (Hint: let $\mathbf{z}_{(ij)}$ differ in sign across j .)

Solution. Take $n_i = 2$, $q_0 = 1$, $z_{i1} = +1$ and $z_{i2} = -1$, so $\eta_{i1} = \eta_{i1}^0 + u_i$ and $\eta_{i2} = \eta_{i2}^0 - u_i$. Then $f(u) = h(\eta_{i1}^0 + u)$ is nondecreasing and $g(u) = h(\eta_{i2}^0 - u)$ is nonincreasing, so the association inequality applies with $-g$ and gives $\text{Cov}\{f(U), g(U)\} \leq 0$. The lemma assumed both functions nondecreasing, which is automatic only when every $z_{(ij)}$ has the same sign — as for an intercept.

40.2.6.3 C. Going deeper**Exercise C1.**

In [Example \(Grunfeld's firms, again\)](#) the mixed-model slopes differ markedly from the independence ones. Add the firm means $\overline{\log \text{value}}_i$ and $\overline{\log \text{capital}}_i$ to the model as extra fixed effects, so that the original covariates carry only within-firm variation. Explain why this makes the random effect orthogonal to the covariates in the population, refit, and compare. What test does the difference between the two specifications suggest?